LKMA: Learnable Kernel and Mamba With Spatial–Spectral Attention Fusion for Hyperspectral Image Classification

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***Abstract*—Transformer models have achieved remarkable success in hyperspectral image classification (HSIC) owing to their strong global modeling capability. However, their quadratic complexity significantly limits their computational efficiency. Recently, Mamba has been applied to HSIC because of its linear complexity, yet it still suffers from an imbalance between global and local modeling. To overcome these challenges, this article proposes a novel learnable kernel and Mamba with spatial–spectral attention fusion (LKMA) framework, which enables the extraction of global–local spatial–spectral features (SSFs) while enhancing edge feature representation. For local feature extraction, the proposed multiscale spatial–spectral feature generator (MSSFG) module captures local SSF by employing multiscale learnable dilation convolutions (LDCs) for spatial features and multiscale dilation convolutions for spectral features. For global feature extraction, a global hidden mixing Mamba (GHMM) module is introduced, which projects hyperspectral image (HSI) features from the feature space to the hidden state space via a hidden state mixing mechanism. This enables the model to capture contextual semantic information and local details from the HSI. To further explore the synergistic effect between spatial and spectral information, the spatial–spectral attention fusion (SSAF) module integrates semantic information across multiple feature groups by combining semantic grouped spatial attention (SGSA) and progressive spectral self-attention (PSSA), enhancing spatial–spectral representations. Extensive experiments demonstrate that the proposed method outperforms state-of-the-art approaches for HSIC.**

***Index Terms*—Global hidden mixing Mamba (GHMM), hyperspectral image classification (HSIC), learnable dilation convolution (LDC), spatial–spectral attention fusion (SSAF).**

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I. Introduction

HYPERSPECTRAL image (HSI) can be seen as 3-D data cubes containing rich spectral features, with hundreds of consecutive bands. Compared with traditional RGB or multispectral images, HSI can reflect both the spatial distribution of features and the spectral characteristics of material components, providing unique advantages for accurately identifying surface targets. In recent years, HSI technology has increasingly been employed in diverse fields, including pollutant identification [1], crop growth monitoring [2], [3], and mineral composition determination [4], [5].

Early studies on hyperspectral image classification (HSIC) primarily relied on machine learning and handcrafted approaches, including support vector machines (SVMs) [6], K-nearest neighbors (KNNs) [7], and random forests (RFs) [8]. However, these methods mainly exploited the spectral features of HSI data, with limited ability to capture spatial features. To better exploit the spatial information contained in HSI data, researchers proposed methods such as extended morphological profiles (EMPs) [9] and sparse manifold representations [10], leading to more accurate classification results. Nevertheless, these approaches depend heavily on predefined parameters and exhibit limited representation capability under complex environmental conditions. In addition, Deng et al. [11] proposed an HSIC method that integrates tensor decomposition-based dictionary learning with relaxed linear regression, achieving superior classification performance.

To address these limitations, deep learning architectures (with powerful nonlinear representation capabilities) have been widely adopted, achieving remarkable success in HSIC and related classification tasks [12], [13], [14], [15], [16]. For instance, Hu et al. [17] proposed the first hierarchical 1-D convolutional neural network (CNN) architecture, which extracts high-level features along the spectral dimension and overcomes the bottleneck of shallow utilization of spectral information in traditional methods. Lee and Kwon [18] constructed a 2-D CNN model for capturing spatial correlations in the HSI and effectively extracted local texture features

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within a single band by multilayer convolutional operations. Hamida et al. [19] proposed a 3-D CNN architecture, which treats the HSI as a 3-D data body and enables joint spectral–spatial feature extraction through a 3-D convolution kernel. Roy et al. [20] further proposed the HybridSN model, which integrates 3-D and 2-D CNN modules, effectively capturing spectral–spatial representations and achieving high classification accuracy with reduced computational overhead. Following this line of research, Gao et al. [21] designed a spatial–spectral perception network for hyperspectral and other images classification, which can obtain more valuable complementary features and further improve classification performance. Similarly, Ren et al. [22] proposed a nonlocal and deep priors method, leveraging self-similarity and global spectral correlations to enhance feature representation. Although CNN-based models perform well in capturing local features, they still exhibit certain limitations in modeling long-range dependencies.

Inspired by the selective attention mechanism of human vision, the Transformer architecture overcomes the limitations of local receptive fields inherent in traditional CNNs by employing global self-attention [23], [24], [25]. Vaswani et al. [26] first proposed the Transformer, which utilizes global self-attention to establish long-range dependencies between sequence elements, significantly enhancing the modeling capability for sequences. Dosovitskiy et al. [27] introduced the vision Transformer (ViT), which pioneered the use of Transformer architectures for the image domain by implementing global modeling through image patch embedding and positional encoding. Hong et al. [28] proposed SpectralFormer, which serializes hyperspectral pixels into tokens and models long-range dependencies in the spectral dimension using multihead self-attention. Zhong et al. [29] further proposed the spectral–spatial Transformer network (SSTN) model, introducing a spatial–spectral dual-branch structure to separately model the interactions between local spatial neighborhoods and global spectral sequences. Roy et al. [30] developed MorphFormer, which innovatively integrates morphological operations into the Transformer architecture, effectively improving the recognition of subtle ground objects. To balance spectral temporal dependencies and global spatial interactions, Zhang et al. [31] proposed the PFS3F model, which fuses superpixel spatial structures with semantic spectral features to enhance boundary recognition. Despite the above methods having demonstrated great advantages in modeling global dependencies, their quadratic complexity poses significant challenges to computational efficiency.

To tackle these computational issues, recent advancements have centered on developing architectures that offer efficient sequence modeling. Among these, the Mamba [32] architecture, built upon state-space models (SSMs), has gained widespread attention in HSIC for its linear complexity and dynamic sequence modeling capabilities. He et al. [33] proposed IGroupSS-Mamba, which improves computational efficiency while preserving long-range dependencies by introducing an interval grouping mechanism. Li et al. [34] constructed MambaHSI, pioneering the use of the Mamba framework for HSIC and demonstrating superior global modeling capabilities compared to Transformer-based methods. Yang et al. [35] introduced the GraphMamba model, which improves spatial structural awareness and computational efficiency by incorporating a global mask and adaptive graph convolution. He et al. [36] proposed HSI-MFormer, which achieves a complementary integration of global context modeling and local feature through the collaborative design of cross-scale Mamba experts (CMEs) and intrascale Transformer experts (ITEs). Although existing Mamba-based methods demonstrate impressive computational efficiency, they still suffer from an imbalance between global and local modeling.

In light of the above challenges, this work proposes a novel framework named learnable kernel and Mamba with spatial–spectral attention fusion (LKMA) for HSIC. The proposed framework comprises three key components: a multiscale spatial–spectral feature generator (MSSFG), a global hidden mixing Mamba (GHMM) module, and a spatial–spectral attention fusion (SSAF) module. LKMA enables the modeling of global and local spatial–spectral features (SSFs) while further enhancing the representation of edge information. Specifically, the MSSFG module is first employed to extract local SSF. Within this module, multiscale learnable dilation convolutions (LDCs) are proposed for spatial feature extraction, while multiscale dilation convolutions are used for spectral feature extraction. Next, the GHMM module leverages a hidden state mixing mechanism to project features into the hidden state space, facilitating the extraction of both local details and contextual semantic information in the HSI. To further enhance SSF representation, the SSAF module is designed by integrating semantic grouped spatial attention (SGSA) and progressive spectral self-attention (PSSA), thereby constructing deep spatial–spectral semantic correlations and improving classification performance.

The main contributions of this work are summarized as follows.

1) A novel LKMA framework is proposed for HSIC, which integrates the extraction of global and local SSF with enhanced edge feature representation.
2) To the best of our knowledge, an LDC method based on a “center-neighbor” comparison mechanism is proposed for the first time, which improves the extraction of edge-salient features in HSI through a parallel architecture.
3) A new GHMM module is constructed based on a hidden state mixing mechanism, which projects HSI data from the feature space to the hidden state space. This design balances global and local receptive fields and facilitates extracting local and contextual information while maintaining linear complexity.
4) To capture the synergistic effect of spatial and spectral features, the SSAF module integrates semantic information across feature groups via SGSA and PSSA, enhancing spatial–spectral representations.

The rest of this article is organized as follows. Section II briefly reviews some relevant algorithms. The proposed method is discussed in Section III. Section IV discusses the experimental results obtained using four benchmark HSI

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datasets. Section V provides a summary of the article and discusses potential directions for further research.

II. Related Work

This section briefly reviews the concepts of self-attention mechanisms, SSMs, and state-space duality (SSD), laying a foundation for the submodules and framework proposed in Section III.

A. Self-Attention Mechanism

The self-attention mechanism serves as a core technique for capturing internal dependencies by dynamically constructing association weights between input features.

Assume that the input feature matrix is $\mathbf{T} \in \mathbb{R}^{H\times W\times F}$, where H denotes the spatial height, W refers to the spatial width, and F corresponds to channel dimensions. The fundamental operation involves applying three distinct linear projections to $\mathbf{T}$, resulting in the query $\mathbf{Q}$, key $\mathbf{K}$, and value matrices $\mathbf{V}$, as shown in the following:

$$\mathbf{Q} = \mathbf{T} \times \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{T} \times \mathbf{W}_K, \quad \mathbf{V} = \mathbf{T} \times \mathbf{W}_V \tag{1}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{F\times d_k}$ are projection weight matrices and d_k is the feature dimension after projection. To measure the input element associations, the attention score is computed to reflect the vector similarity by the dot product of $\mathbf{Q}$ and $\mathbf{K}$. The scaling factor d_k is introduced to balance the gradient, yielding a scaled attention score, which can be expressed as

$$\text{Score} = \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}. \tag{2}$$

Then, (2) is normalized to generate the attention weights, which are applied to weight and sum the value matrix $\mathbf{V}$ for computing the final output $\mathbf{O}$. The output calculation is given by

$$\mathbf{O} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}. \tag{3}$$

This output captures long-range dependencies and contextual information, enhancing feature representation and improving task performance.

B. State-Space Models

An SSM is a sequence model, derived from continuous-time linear time-invariant (LTI) systems, which maps an input sequence $x(t) \in \mathbb{R}$ into an output sequence $y(t) \in \mathbb{R}$. In continuous-time scenarios, the transformation is described by a linear ordinary differential equation

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{4}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are the projection matrices, N is the number of states, and $h(t) \in \mathbb{R}^{N\times 1}$ is a hidden state. To adapt the discrete data in deep learning, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ are transformed via zero-order hold (ZOH), producing the discrete-time system

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$$

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\cdot\Delta\mathbf{B} \tag{5}$$

where Δ stands for the time-scale parameter and $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ represent the discrete matrices obtained by discretizing the continuous matrices $\mathbf{A}$ and $\mathbf{B}$, respectively.

After discretization using the ZOH method, (4) is transformed into the following recursive form:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{6}$$

The discretized matrices $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ within recursive equations empower the linear time-variant discrete system to process sequential data at each t, adaptively capturing features from the hidden state.

C. State-Space Duality

Standard SSMs capture sequential dependencies via recursive state transitions, but they can be computationally intensive and less scalable for long sequences. To address these limitations, the recently proposed SSD reformulates SSMs for improved optimization. In Mamba2 [37], SSD reduces the evolution matrix $\mathbf{A}$ to scalar form and discretizes it to obtain $\mathbf{a} \in \mathbb{R}^L$. Based on this, SSD reformulates (4) as the following matrix transformation:

$$\mathbf{y} = \text{SSD}(\mathbf{x}, \mathbf{a}, \mathbf{B}, \mathbf{C}) = \left(\mathbf{M} \odot \left(\mathbf{C}\mathbf{B}^T\right)\right)\mathbf{x}$$

$$\mathbf{M}_{mn} = \begin{cases} \prod_{k=n+1}^{m} \mathbf{a}_k, & \text{if } m > n \\ 1, & \text{if } m = n \\ 0, & \text{if } m < n \end{cases} \tag{7}$$

where $\mathbf{x} \in \mathbb{R}^{L\times D}$ is the input sequence (sequence length L and feature dimension D), $\mathbf{a} = (a_1, \ldots, a_L)$ with a_k being its kth element, $\mathbf{M} \in \mathbb{R}^{L\times L}$ is a lower-triangular matrix capturing temporal dependencies, and $\odot$ denotes the Hadamard product.

III. Proposed Method

This section presents the overall framework of the LKMA for HSIC, followed by detailed descriptions of its key components: the MSSFG module for extracting spatial and spectral features (with an LDC module for spatial feature extraction), a GHMM module for capturing long-range dependencies, and an SSAF module for fusing spatial and spectral features.

A. Overall Framework

Fig. 1(a) shows the complete framework of LKMA for HSIC. It is composed of an MSSFG, a GHMM, an SSAF, and a prediction module.

Assume that the original HSI data is a 3-D tensor $\mathbf{X} \in \mathbb{R}^{M\times H\times W}$, where M denotes the total number of spectral bands, and H and W correspond to the spatial height and width, respectively. To preserve the main spectral information while reducing the dimensionality, principal component analysis (PCA) [38] is initially employed to obtain a lower-dimensional representation, which can be formulated as

$$\mathbf{X}_p = \text{PCA}(\mathbf{X}) \tag{8}$$

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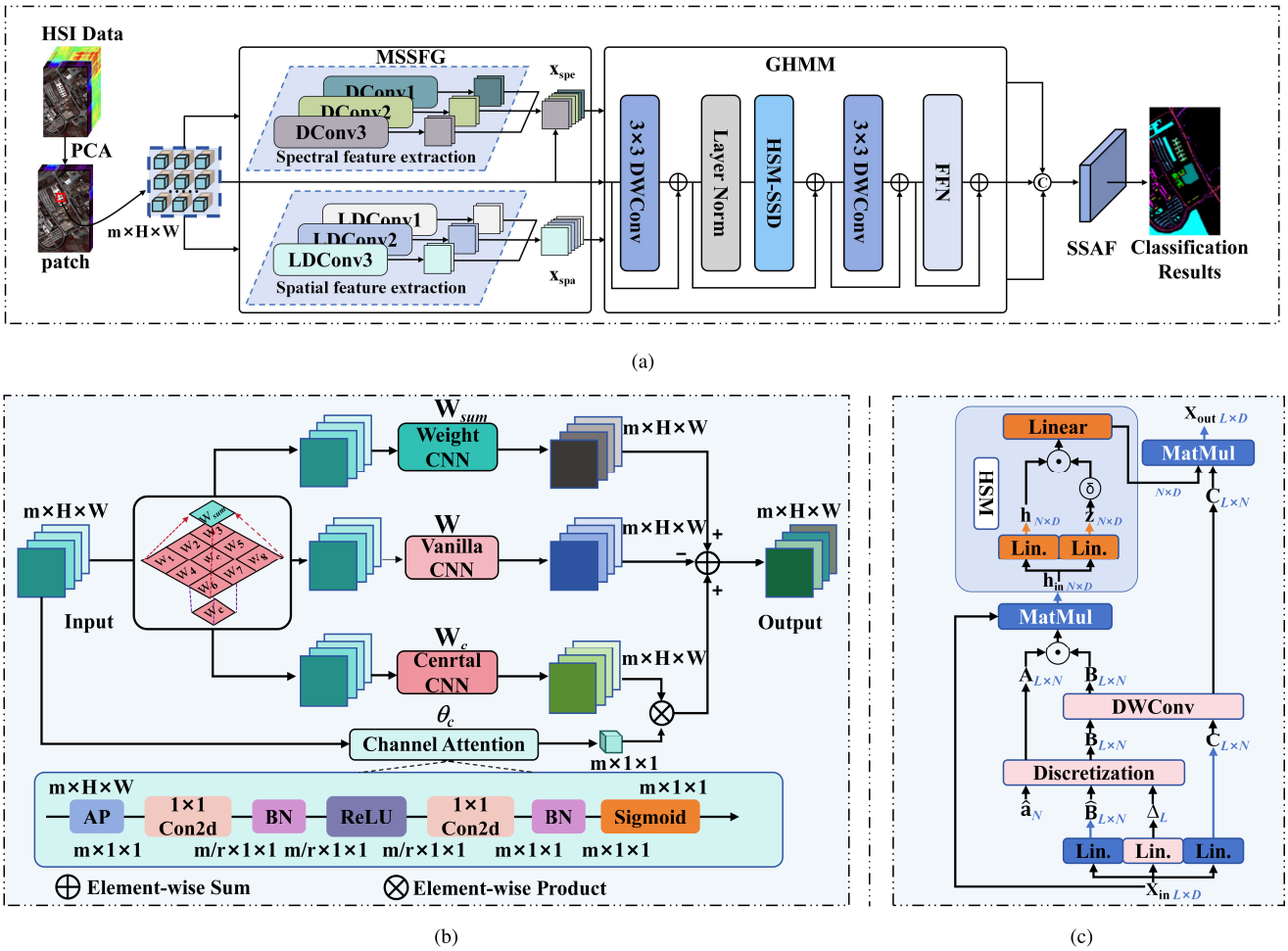


Fig. 1. (a) Overall framework of the LKMA for HSIC. (b) Computational procedure of the proposed LDC. (c) Structural flow of the proposed HSM-SSD.

where $\mathbf{X}_p \in \mathbb{R}^{m\times H\times W}$ denotes the reduced feature representation and m corresponds to the spectral dimension following dimensionality reduction.

The reduced feature $\mathbf{X}_p$ is subsequently passed through the MSSFG module to capture spatial and spectral features (SSFs). The spectral branch extracts the correlations among spectral bands using multiscale dilation convolutions (DConv_j), while the spatial branch extracts spatial multiscale features through three sets of LDCs (LDConv_j). The corresponding formulation is given as follows:

$$\mathbf{X}_{\mathrm{spe}} = \sum_{j=1}^{3} \mathrm{DConv}_j\left(\mathbf{X}_p\right) + \mathbf{X}_p \tag{9}$$

$$\mathbf{X}_{\mathrm{spa}} = \mathrm{concat}\begin{pmatrix} \mathrm{LDConv}_1(\mathbf{X}_p) \\ \mathrm{LDConv}_2(\mathbf{X}_p) \\ \mathrm{LDConv}_3(\mathbf{X}_p) \end{pmatrix} \tag{10}$$

where $\mathbf{X}_{\mathrm{spe}}$ and $\mathbf{X}_{\mathrm{spa}} \in \mathbb{R}^{H\times W\times K}$ represent the spectral and spatial features. The feature channel dimension is set as $K = 48$. DConv_j $(j = 1, 2, 3)$ corresponds to convolution kernel sizes of $5\times1\times1$, $7\times1\times1$, and $9\times1\times1$ with dilation rates of 1, 2, and 4, and LDConv_j $(j = 1, 2, 3)$ corresponds to convolution operations using kernels sized 3×3, 5×5, and 7×7, all with a dilation rate of 2.

Subsequently, the extracted features are delivered to the GHMM module for capturing long-range dependencies, resulting in $\bar{\mathbf{X}}_{\mathrm{spe}}$ and $\bar{\mathbf{X}}_{\mathrm{spa}}$. To retain the original global context, a global skip connection is applied. Then, SSAF integrates all features, and the final prediction is obtained through the classification head. The process is formulated as

$$\begin{aligned} \bar{\mathbf{X}}_{\mathrm{spe}} &= \mathrm{GHMM}\left(\mathbf{X}_{\mathrm{spe}}\right), \quad \bar{\mathbf{X}}_{\mathrm{spa}} = \mathrm{GHMM}\left(\mathbf{X}_{\mathrm{spa}}\right) \\ \mathbf{X}_{\mathrm{fusion}} &= \mathrm{SSAF}\left(\mathrm{concat}\left(\bar{\mathbf{X}}_{\mathrm{spe}}, \bar{\mathbf{X}}_{\mathrm{spa}}, \mathrm{GHMM}\left(\mathbf{X}_p\right)\right)\right) \\ pred &= \mathrm{Classifier}\left(\mathrm{avg}\left(\mathbf{X}_{\mathrm{fusion}}\right)\right) \end{aligned} \tag{11}$$

where GHMM$(\cdot)$ denotes the GHMM module, SSAF$(\cdot)$ is the SSAF module, avg$(\cdot)$ represents average pooling, and Classifier$(\cdot)$ refers to the classification head. Through the above operations, the spatial–spectral correlations of HSI can be effectively modeled.

B. Learnable Dilation Convolution

LDC is a core component of MSSFG for spatial feature extraction, which uses the "center-neighborhood" comparison mechanism of local saliency kernels, as shown in Fig. 1(b). It integrates learnable weights with a channel attention mechanism and applies dilation to expand the receptive field while maintaining parameter efficiency.

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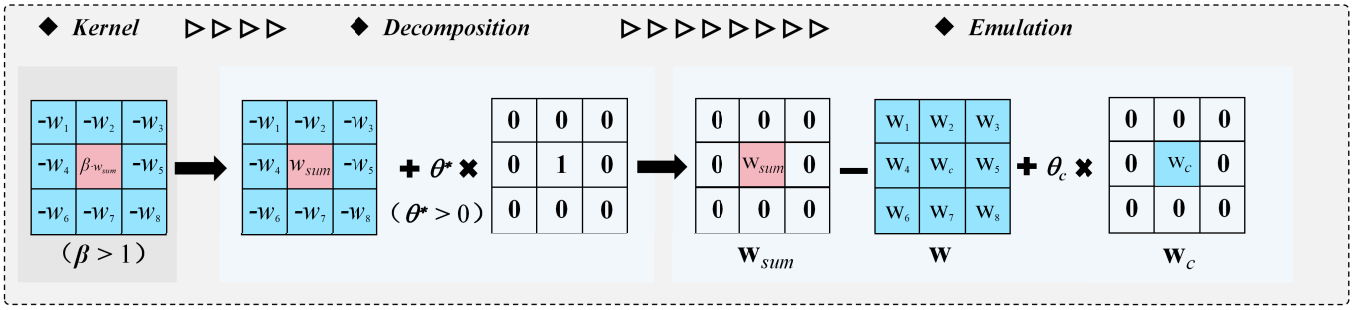


Fig. 2. Decomposition method of kernel dilation convolution and the structure of its process can be learned, here using a 3 × 3 kernel as an example.

In general, the local saliency kernel $O(\cdot)$ is characterized by the center sum weight w_{sum}. To enhance the saliency of the central feature, the response to the target is amplified by an enhancement factor β ($\beta > 1$), which accentuates the difference between the pixel at the center and its adjacent pixels through dynamic weight adjustment.

Taking a 3 × 3 kernel as an example, as shown in Fig. 2, w_{sum} maintains a defined relationship with the neighborhood weights w_i ($i = 1, 2, \ldots, 8$), which can be expressed as

$$w_{\text{sum}} = \sum_{i=1}^{8} w_i. \tag{12}$$

To ensure that learnability is achieved, for $k \times k$ convolutional kernels, the convolutional kernels can be rearranged in a weight map $\mathbf{W} = \{w_1, \ldots, w_c, \ldots, w_{k^2-1}\}$, where $c = \lceil k^2/2 \rceil$, w_c is the center weight, and $w_i (i \in \{1, \ldots, k^2 - 1\})$ is the neighborhood weight.

To prevent feature loss, an enhanced saliency learnable kernel is introduced. As shown in Fig. 2, for an input $\mathbf{F} \in \mathbb{R}^{H \times W}$, the enhanced kernel $\mathbf{F}^*$ is mathematically represented by

$$\mathbf{F}^* = O(\mathbf{F}) + \boldsymbol{\theta}^*(\mathbf{F}) \tag{13}$$

where $O(\mathbf{F})$ represents the local saliency kernel applied to the input feature $\mathbf{F}$ and $\theta^*(\cdot)$ is an extended mapping of the enhancement factor $\mathbf{F}^*$; then, a channel learnable kernel $\mathbf{W}_s$ can be expressed as

$$\mathbf{W}_s = \mathbf{W}_{\text{sum}} - \mathbf{W} + \theta_c \cdot \mathbf{W}_c \tag{14}$$

where $\mathbf{W}_{\text{sum}}$ and $\mathbf{W}_c$ are extensions of the $k \times k$ mapping of w_{sum} and w_c, and in addition, the value domain of θ_c is $[0, 1]$.

Given an input $\mathbf{X_p} \in \mathbb{R}^{m \times H \times W}$, we augment the features using a salient channel attention module $\theta_c(\cdot)$, as shown in Fig. 1(b), which combines a global average pool (AP) to perceive the overall features. The sigmoid layer within the channel attention module constrains the significant representation of neighbors in $[0, 1]$, which is consistent with the range of values of θ_c in the kernel function. The output produced by the learnable kernel can be formulated as

$$\mathbf{Y} = \mathbf{W}_{\text{sum}}(\mathbf{X_p}) - \mathbf{W}(\mathbf{X_p}) + \theta_c(\mathbf{X_p}) \cdot \mathbf{W}_c(\mathbf{X_p}) \tag{15}$$

where the shape of $\mathbf{Y}$ is aligned with the input; this kernel learns the salient features of the aggregation through a convolution operation.

C. Global Hidden Mixing Mamba

To capture the global contextual semantic information between SSF, we design a GHMM module centered on a hidden state mixer-based SSD (HSM-SSD), and its structure is shown in Fig. 1(a).

Specifically, the core idea of HSM-SSD is to map the input features to the hidden state space, thereby achieving feature mixing and augmentation. The structure of HSM-SSD is depicted in Fig. 1(c). To adapt to sequence modeling in HSM-SSD, the spatial dimension is first processed to obtain the input sequence $\mathbf{X}_{\text{in}} \in \mathbb{R}^{L \times D}$, where $L = H \times W$ denotes the sequence length and D corresponds to the channel dimension. To capture dynamic properties within input feature $\mathbf{X}_{\text{in}}$, it is first dimensionally expanded by linear projection to generate three matrices $\hat{\mathbf{B}}, \mathbf{C}, \boldsymbol{\Delta} \in \mathbb{R}^{L \times N}$, where N specifies the amount of states. The projection step proceeds as

$$\hat{\mathbf{B}}, \mathbf{C}, \boldsymbol{\Delta} = \text{Linear}(\mathbf{X}_{\text{in}}). \tag{16}$$

To enhance the interaction capability of local positions, depthwise separable convolutions are applied to $\hat{\mathbf{B}}$ and $\mathbf{C}$. Subsequently, a discretization operation, implemented using the standard ZOH method, is performed on the core learnable parameters $\hat{\mathbf{a}} \in \mathbb{R}^N$ (representing the learnable parameter of the control state transfer), $\hat{\mathbf{B}}$, and $\boldsymbol{\Delta}$ to generate $\mathbf{A}$ and $\mathbf{B}$ for sequence modeling. The following equation represents this process:

$$\begin{aligned} \hat{\mathbf{B}}, \mathbf{C} &= \text{DWConv}(\hat{\mathbf{B}}, \mathbf{C}) \\ \mathbf{A}, \mathbf{B} &= \text{Discretization}(\hat{\mathbf{a}}, \hat{\mathbf{B}}, \boldsymbol{\Delta}). \end{aligned} \tag{17}$$

Then, to achieve global dependency capture and optimize feature representation, the initial hidden state $\mathbf{h}_{\text{in}}$ is computed via matrix interaction. $\mathbf{h}_{\text{in}}$ is linearly projected to adapt to the subsequent gating mechanism and split into two feature components $\mathbf{h}, \mathbf{z} \in \mathbb{R}^{N \times D}$. This operation linearly maps the high-dimensional features of $\mathbf{h}_{\text{in}}$ into a more flexible feature space, where $\mathbf{h}$ is used for messaging and $\mathbf{z}$ is used for gating control, respectively. Followed by the nonlinear adjustment of the gating mechanism, the operation is formally described as

$$\begin{aligned} \mathbf{h}_{\text{in}} &= (\mathbf{A} \odot \mathbf{B})^T \mathbf{X}_{\text{in}} \\ \mathbf{h}, \mathbf{z} &= \text{Linear}(\mathbf{h}_{\text{in}}) \\ \mathbf{h} &= \text{Linear}(\mathbf{h} \odot \sigma(\mathbf{z})) \end{aligned} \tag{18}$$

where $\sigma(\cdot)$ stands for the sigmoid activation function.

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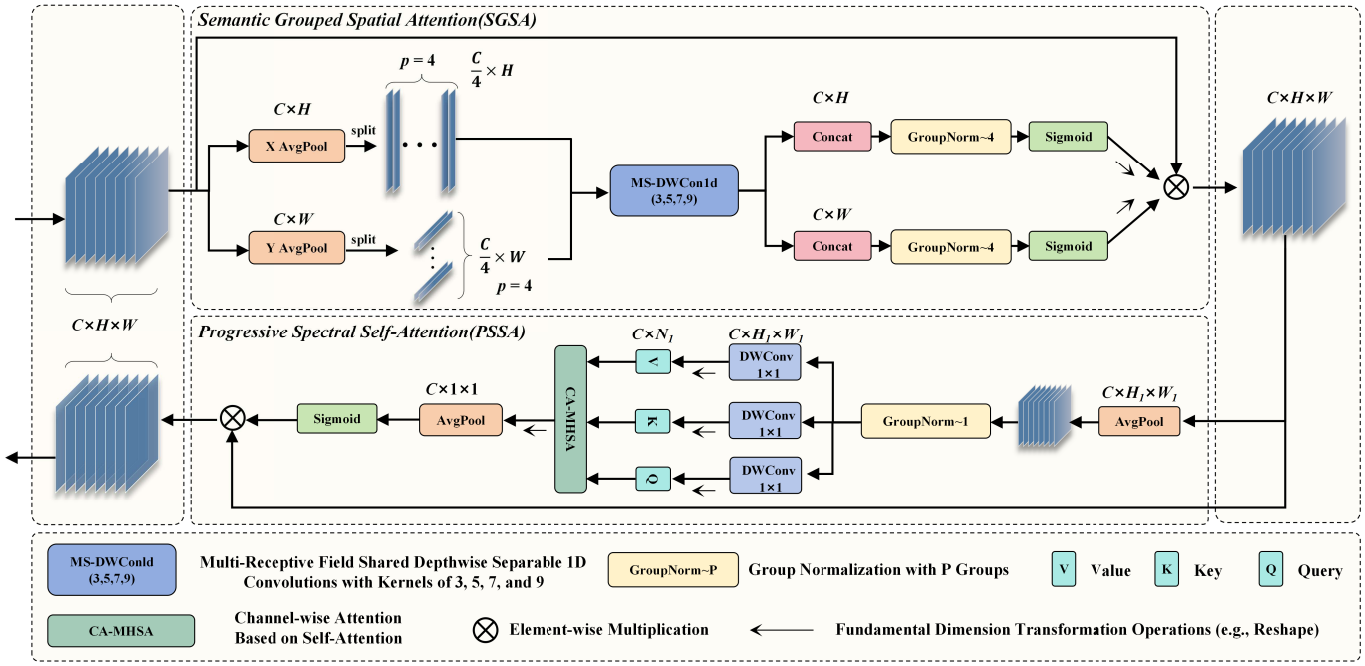


Fig. 3. SSAF module integrates SGSA and PSSA to fuse SSF, enhancing feature representations. p denotes the group count used to divide the subfeatures. SGSA aggregates horizontal and vertical feature groups to strengthen spatial information, while PSSA progressively refines spectral semantics.

After completing the hidden state computation and gating adjustment, HSM-SSD maps the refined $\mathbf{h}$ back to the original feature space dimension through the output mapping matrix C. It can be expressed as

$$\mathbf{X}_{\text{out}} = \mathbf{Ch}. \tag{19}$$

Through the above operations, the module ensures feature dimension compatibility and achieves the extraction of both local and contextual information with linear complexity.

D. Spatial–Spectral Attention Fusion

To capture the synergy between SSF in HSI, an SSAF module is designed, as illustrated in Fig. 3. This module constructs an SSF fusion mechanism through the collaboration between SGSA and PSSA.

Given an input feature $\mathbf{X}_1 \in \mathbb{R}^{C\times H\times W}$, it is first decomposed into horizontal feature $\mathbf{X}_H \in \mathbb{R}^{C\times W}$ and vertical feature $\mathbf{X}_W \in \mathbb{R}^{C\times H}$ via performing average pooling separately on the height and width. To capture varying spatial feature distributions, these features are further divided into four groups of independent subfeatures $\mathbf{X}_H^p$ and $\mathbf{X}_W^p$ $(p = 1, 2, 3, 4)$, each with $C/4$ channels. The decomposition step can be represented as follows:

$$\begin{aligned}\mathbf{X}_H^p &= \mathbf{X}_H\left[(p-1)\times\frac{C}{4} : p\times\frac{C}{4}, :\right]\\ \mathbf{X}_W^p &= \mathbf{X}_W\left[(p-1)\times\frac{C}{4} : p\times\frac{C}{4}, :\right].\end{aligned} \tag{20}$$

To further enrich the semantic information, multiscale depth-separable 1-D convolution (with kernel sizes of 3, 5, 7, and 9) is used for each group of subfeatures to capture the information at different scales, with the operations defined as

$$\begin{aligned}\tilde{\mathbf{X}}_H^p &= \text{DWConv1d}_{k_p}\left(\mathbf{X}_H^p\right)\\ \tilde{\mathbf{X}}_W^p &= \text{DWConv1d}_{k_p}\left(\mathbf{X}_W^p\right)\end{aligned} \tag{21}$$

where $\tilde{\mathbf{X}}_H^p$ and $\tilde{\mathbf{X}}_W^p$ represent spatial features extracted from the pth subfeature extracted along horizontal and vertical dimensions, respectively. k_p denotes the kernel size applied to the pth subfeature.

Subsequently, the features are normalized by group normalization (GN) and activated by the Sigmoid function to generate spatial attention maps Attn_H and Attn_W. The spatially enhanced feature $\mathbf{X}_s$ is then obtained via elementwise multiplication, as formulated in the following:

$$\begin{aligned}\text{Attn}_H &= \sigma\left(\text{GN}_H\left(\sum_{p=1}^{4}\tilde{\mathbf{X}}_H^p\right)\right)\\ \text{Attn}_W &= \sigma\left(\text{GN}_W\left(\sum_{p=1}^{4}\tilde{\mathbf{X}}_W^p\right)\right)\\ \mathbf{X}_s &= \text{Attn}_H\times\text{Attn}_W\times\mathbf{X}_1\end{aligned} \tag{22}$$

where $\text{GN}_H(\cdot)$ and $\text{GN}_W(\cdot)$ refer to GN performed along the height and width axes, respectively.

In addition, this study introduces a PSSA module aimed at enhancing semantic information along the spectral dimension. In detail, adaptive average pooling employing a kernel sized 7×7 operates on $\mathbf{X}_s$ to obtain the pooled feature $\mathbf{X}_a \in \mathbb{R}^{C\times H_1\times W_1}$. Then, the projection function generates $\tilde{\mathbf{Q}}$, $\tilde{\mathbf{K}}$, and $\tilde{\mathbf{V}}$ from $\mathbf{X}_a$. The spectral attention $\mathbf{X}_{\text{attn}}$ is computed to capture dependencies between spectral channels. The obtained attention map undergoes a sigmoid function and is fused

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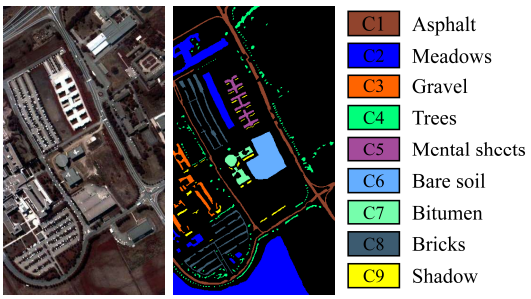


Fig. 4. PU dataset. (a) False-color map. (b) Ground truth. (c) Legends.

with $\mathbf{X}_s$ via elementwise multiplication to refine the spectral semantics and enhance the model's capability in characterizing spectral information. The calculations are outlined as

$$\tilde{\mathbf{Q}} = F^{Q}_{\text{proj}}(\mathbf{X}_a),\ \tilde{\mathbf{K}} = F^{K}_{\text{proj}}(\mathbf{X}_a),\ \tilde{\mathbf{V}} = F^{V}_{\text{proj}}(\mathbf{X}_a)$$
$$\mathbf{X}_{\text{attn}} = \text{Attn}\left(\tilde{\mathbf{Q}}, \tilde{\mathbf{K}}, \tilde{\mathbf{V}}\right) = \text{Softmax}\left(\frac{\tilde{\mathbf{Q}}\tilde{\mathbf{K}}^{T}}{\sqrt{C}}\right)\tilde{\mathbf{V}}$$
$$\mathbf{X}_c = \mathbf{X}_s \odot \sigma\left(\text{Pool}_{(H_1,W_1)\to(1,1)}(\mathbf{X}_{\text{attn}})\right) \tag{23}$$

where $F_{\text{proj}}(\cdot)$ denotes the mapping function used to generate the projected features $\tilde{\mathbf{Q}}, \tilde{\mathbf{K}}, \tilde{\mathbf{V}} \in \mathbb{R}^{C\times N_1}$ ($N_1 = H \times W$). The resulting output $\mathbf{X}_c$ is the spatial feature $\mathbf{X}_s$ refined by the learned spectral attention weights.

Finally, a serial architecture is adopted to integrate the SGSA and PSSA modules, achieving SSAF, which is formally defined as

$$\text{SSAF}(\mathbf{X}_1) = \text{PSSA}(\text{SGSA}(\mathbf{X}_1)). \tag{24}$$

IV. Experiments

A. Datasets Description

1) Pavia University: The Pavia University (PU) dataset was recorded using the Reflective Optics Spectrographic Imaging System (ROSIS-3) sensor over Pavia, Italy. It comprises an HSI of size 610 × 340 pixels across 103 spectral bands. The image is classified into nine categories, totaling 42 776 labeled samples, such as asphalt, grass, and buildings. The false-color map and ground-truth map are shown in Fig. 4.

2) Salinas: The Salinas (SA) dataset was captured in Salinas Valley, CA, USA, using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) hyperspectral sensor. The original data contain 224 spectral bands, with 204 bands remaining after discarding those adversely affected by water absorption. The image measures 512 × 217 pixels and is categorized into 16 classes, totaling 54 129 labeled samples, including categories such as cropland and celery. The false-color map and ground-truth map are shown in Fig. 5.

3) Fanglu Tea Farm (Tea Farm): The Tea Farm dataset was collected using a pushbroom hyperspectral imager at a tea plantation in Fangruzhi Village, Changzhou, Jiangsu, China. It features an image of 348 × 512 pixels with 80 spectral bands, and the dataset package covers ten different categories, totaling 53 734 labeled samples, such as Sargasso Pine and Bamboo Forest. The false-color map and ground-truth map are shown in Fig. 6.



Fig. 5. SA dataset. (a) False-color map. (b) Ground truth. (c) Legends.

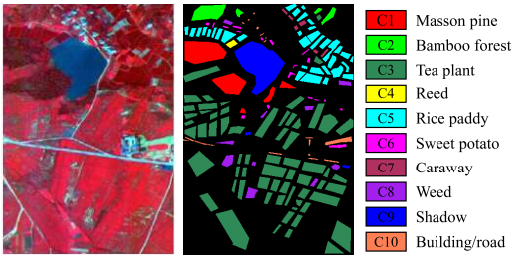


Fig. 6. Tea Farm dataset. (a) False-color map. (b) Ground truth. (c) Legends.

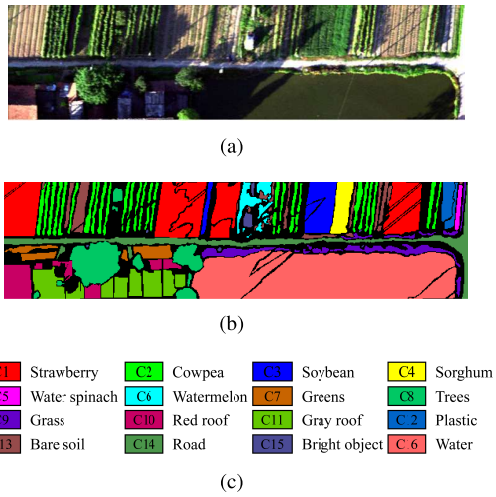


Fig. 7. HanChuan dataset. (a) False-color map. (b) Ground truth. (c) Legends.

4) WHU-Hi-HanChuan (HanChuan): The HanChuan dataset was collected in Hanchuan, Hubei, China. The image comprises 1217 × 303 pixels and contains 274 spectral bands. The dataset covers 16 different categories, amounting to 2 57 530 labeled samples, including categories such as strawberry and cowpea. The false-color map and ground-truth map are shown in Fig. 7.

B. Experimental Settings

1) Evaluation Metrics: Following existing HSIC studies, the overall accuracy (OA), average accuracy (AA), and kappa

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coefficient (Kappa) are adopted as evaluation metrics [39], [40], [41].

2) Comparison Methods: To validate the effectiveness of the proposed LKMA method, four types of HSIC architectures were selected for comparison, including the traditional machine learning method (SVM [6]), CNN-based methods (HybridSN [20] and SPRN [42]), Transformer-based methods (HMSSF [43], MASSFormer [44], and GSC-VIT [45]), and Mamba-based methods (3DSS-Mamba [46] and IGroupSS-Mamba [33]).

a) SVM: This method is based on statistical learning theory and achieves classification by constructing an optimal classification hyperplane.

b) HybridSN: This method combines 3-D convolution with 2-D convolution to extract SSF and reduces computational complexity by lightening the layer structure.

c) SPRN: This method divides spectral subbands by group convolution, combines parallel residual blocks that facilitate hierarchical feature learning across multiple scales, and introduces a homogeneous pixel detection module to suppress large-size patch interference.

d) HMSSF: This method employs a two-branch architecture with a spatial–spectral token generation module (SSTG) and a multiscale self-attention (MSSA) module to model multiscale spatial–spectral dependencies, followed by feature fusion using a spatial–spectral attention aggregation (SSAA) module.

e) MASSFormer: This method introduces the memory token mechanism (MT) and memory-enhanced encoder (MATE) to store prior knowledge through memory tokens to alleviate the information decay problem of deep networks and realize long-range dependency modeling.

f) GSC-VIT: This method uses a lightweight architecture of grouped separable convolution and ViT to balance global–local features and reduce the burden of high-dimensional computation through spectral calibration blocks and grouped separable ViT modules.

g) 3DSS-Mamba: This method introduces a 3-D selective scanning mechanism combined with the S6 model, modeling global long-range dependencies, balancing linear efficiency with high-dimensional modeling.

h) IGroupSS-Mamba: This method proposes the interval group S6 mechanism (IGSM), which groups high-dimensional features in intervals for unidirectional S6 scanning. It extracts global features by an interval group spatial–spectral block (IGSSB) and integrates downsampling of pixel aggregation to realize multiscale processing.

3) Implementation Details: The experiments were all conducted on the PyTorch platform, and to eliminate the randomness of the experiments and the effect of parameter initialization, all the experiments obtained were averaged over ten consecutive runs. For our method, 100 epochs were run for training on all datasets. The parameters are set as follows: the PCA, the batch size, the patch size, and the learning rate for the PU dataset are 15, 512, 11, and $5e^{-}4$, respectively. For the SA dataset, they are 20, 512, 17, and $1e^{-}3$. For the Tea Farm dataset, they are 25, 112, 11, and $3e^{-}4$. For the HanChuan dataset, they are 20, 64, 17, and $1e^{-}3$. All runs were performed on the same hardware: an NVIDIA GeForce RTX 4060 Ti GPU, ×64 Intel[1] Core[2] i7-14700KF CPU, and 128-GB RAM.

C. Parameter Analysis

1) Analysis of the PCA: While keeping all other parameters constant, we investigated the effect of different PCA dimensions on classification performance with candidate values {10, 15, 20, 25, 30}. As shown in Fig. 8(a), for the PU dataset, OA increases initially and then decreases as the PCA dimension grows, suggesting that too many components may introduce redundant information. The SA dataset shows a slight increase in OA, reaching a peak at moderate PCA dimensions before slightly declining at larger dimensions. In the Tea Farm dataset, OA generally rises with increasing PCA dimensions and then slightly declines after reaching the highest performance. For the HanChuan dataset, OA also increases at first and then gradually decreases for larger dimensions. Overall, the optimal PCA dimension varies across datasets.

2) Analysis of the Patch Size: To investigate the impact of patch size on classification performance, we kept all other parameters constant and considered a candidate set of values {11, 13, 15, 17, 19}. As shown in Fig. 8(b), the PU and Tea Farm datasets generally perform better with smaller patch sizes, whereas the SA and HanChuan datasets show improved performance with larger patch sizes. These observations highlight the importance of selecting an appropriate patch size for each dataset to achieve a proper balance between local feature extraction and classification performance.

3) Analysis of the Learning Rate: The learning rate is a crucial parameter in model training, directly affecting training stability and classification performance. We considered a candidate set of values $\{3e^{-}4, 5e^{-}4, 7e^{-}4, 1e^{-}3, 1.2e^{-}3\}$. Fig. 8(c) shows the OA of the model across different learning rates for the four datasets. The results indicate that the optimal learning rate varies across datasets: some datasets achieve the best performance with smaller learning rates, while others benefit from moderate values. Extremely small or large learning rates generally lead to slightly decreased accuracy or less stable training. This highlights that appropriate adjustment of the learning rate is important for maintaining stable training and strong classification performance across datasets.

D. Comparison With SOTAs

The quantitative analysis [OA(%), AA(%), and Kappa(%)] of the PU, SA, Tea Farm, and HanChuan datasets is summarized in Tables I–IV, with the best results highlighted in bold. The corresponding classification diagrams are shown in Figs. 9–12.

1) PU Dataset: The experiments are conducted on the PU dataset using 0.5% of the available labeled samples for training. The corresponding quantitative classification results and maps are shown in Table I and Fig. 9. According to Table I, the proposed LKMA demonstrates the highest classification performance among traditional, CNN-based,

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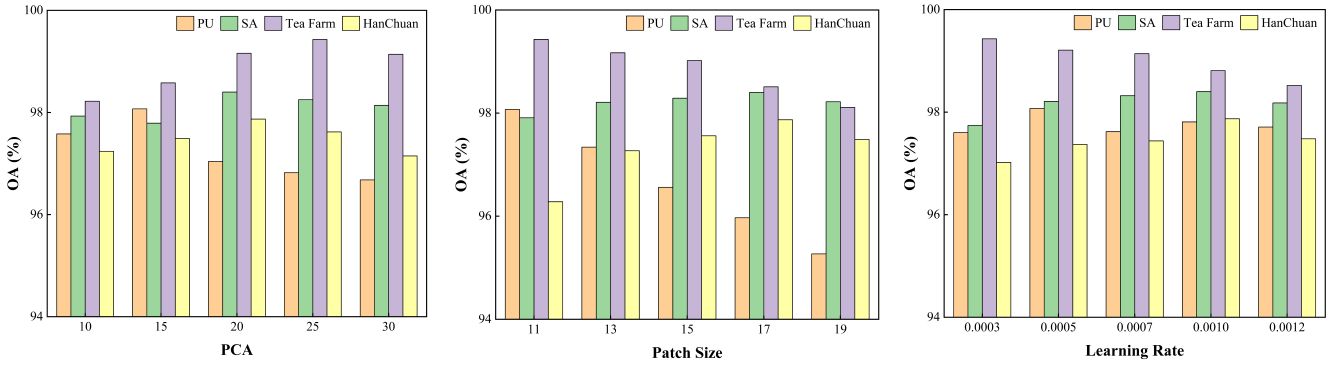


Fig. 8. Analysis of different parameters on OA across PU, SA, Tea Farm, and HanChuan. (a) Effect of PCA. (b) Effect of patch size. (c) Effect of learning rate.

TABLE I
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE PU DATASET USING 0.5% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods		LKMA
	SVM	HybridSN	SPRN	HMSSF	MASSFormer	GSC-VIT	3DSS-Mamba	IGroupSS-Mamba	
C1	86.61	90.43	96.35	97.71	99.06	96.93	87.16	98.52	**99.82**
C2	95.31	98.89	99.08	99.81	99.86	98.84	95.04	99.80	**99.87**
C3	45.46	67.41	83.07	83.74	82.77	82.58	71.84	75.57	**91.49**
C4	78.85	86.40	92.14	81.57	82.81	**94.08**	70.03	81.44	89.93
C5	98.40	99.25	99.05	99.96	99.93	99.91	96.55	**100.00**	**100.00**
C6	61.88	94.53	98.31	97.94	97.38	93.89	68.45	97.75	**99.91**
C7	37.67	82.63	90.06	99.28	99.30	89.43	65.41	99.58	**99.78**
C8	80.20	78.89	93.33	84.43	87.43	94.23	82.48	91.15	**95.00**
C9	**99.76**	72.12	96.72	86.38	91.25	98.43	72.65	94.07	88.30
OA(%)	83.51±1.02	91.82±1.52	96.08±0.58	95.54±0.78	96.11±0.51	96.16±0.72	85.31±6.90	95.98±0.29	**98.07±0.26**
AA(%)	76.02±3.13	85.62±2.63	94.23±1.07	92.31±1.43	93.31±0.77	94.26±1.41	78.85±7.87	93.10±0.60	**96.01±0.56**
Kappa(%)	77.71±1.49	89.10±2.02	94.83±0.72	90.05±1.04	94.82±0.68	96.90±0.95	80.46±9.10	94.65±0.38	**97.44±0.34**

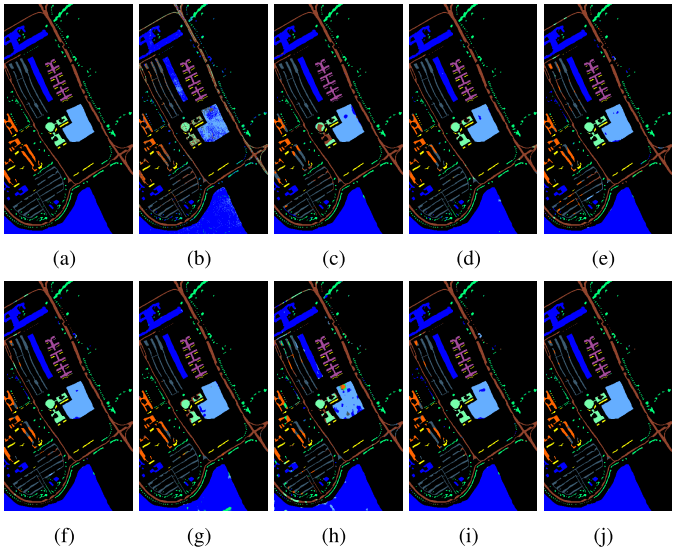


Fig. 9. Classification maps of the PU dataset obtained by different methods. (a) Ground truth. (b) SVM. (c) HybridSN. (d) SPRN. (e) HMSSF. (f) MASSFormer. (g) GSC-VIT. (h) 3DSS-Mamba. (i) IGroupSS-Mamba. (j) LKMA.

Transformer-based, and Mamba-based approaches. Relative to the second-best method, GSC-VIT, LKMA achieves notable improvements of 1.90%, 1.75%, and 2.54% in terms of OA, AA, and Kappa, respectively. Traditional SVM methods are limited in performance because they ignore spatial information. By contrast, CNN-based methods, such as HybridSN, are limited by local receptive fields and difficult to model long-distance dependencies, and Transformer-based methods, such as HMSSF, capture long-distance dependencies through the attentional mechanism but are limited by the complexity of the secondary computation. Mamba-based methods, such as IGroupSS-Mamba, utilize linear computational complexity to achieve long-range modeling but have limited multiscale feature extraction and spatial–spectral interaction. Unlike the above approaches, LKMA effectively extracts multiscale SSF using MSSFG, captures global dependencies with GHMM, and strengthens the interaction between spatial and spectral dimensions via SSAF. The visualization results indicate that LKMA reduces noise artifacts, producing clearer boundary delineation.

2) SA Dataset: The experiments on the SA dataset are conducted using 0.5% of the available labeled samples for training. The corresponding quantitative classification results and maps are shown in Table II and Fig. 10. LKMA significantly improves classification performance by effectively integrating multiscale SSF extraction, global

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TABLE II
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE SA DATASET USING 0.5% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods		LKMA
	SVM	HybridSN	SPRN	HMSSF	MASSFormer	GSC-VIT	3DSS-Mamba	IGroupSS-Mamba	
C1	96.25	99.98	99.98	99.90	99.95	96.63	95.82	**100.00**	99.91
C2	98.19	99.92	**100.00**	**100.00**	**100.00**	99.35	97.65	**100.00**	**100.00**
C3	83.61	93.47	95.38	99.95	**100.00**	89.42	96.23	**100.00**	**100.00**
C4	**98.21**	97.02	98.13	95.81	97.14	95.49	89.13	97.58	95.64
C5	96.23	93.96	96.00	**100.00**	99.97	92.73	97.81	**100.00**	99.86
C6	98.47	99.71	99.95	99.94	99.89	**99.96**	97.75	99.88	99.40
C7	98.86	99.71	**100.00**	99.86	99.94	99.97	92.72	99.87	99.72
C8	87.74	94.60	91.43	94.57	93.99	89.36	89.11	94.70	**99.36**
C9	97.81	99.90	**100.00**	**100.00**	**100.00**	99.82	98.94	**100.00**	**100.00**
C10	84.65	97.48	95.45	98.84	98.88	93.78	92.32	98.73	**99.84**
C11	85.83	98.92	92.59	**99.77**	99.67	83.98	84.66	99.71	97.98
C12	99.48	99.07	99.17	99.77	99.93	97.11	83.71	**99.85**	99.42
C13	97.50	90.85	**99.64**	92.36	92.65	98.84	75.14	91.04	89.46
C14	90.43	97.21	**99.31**	96.20	97.14	97.20	81.75	95.64	96.03
C15	52.31	**92.96**	89.86	89.93	88.91	82.48	80.36	91.21	92.87
C16	83.65	97.32	96.88	99.25	99.97	92.92	86.44	99.52	**100.00**
OA(%)	87.50±1.23	96.75±0.85	95.86±0.77	97.08±0.40	96.81±0.28	93.27±1.04	90.90±1.32	97.29±0.32	**98.40±0.22**
AA(%)	90.58±1.66	97.01±0.69	97.11±0.77	97.88±0.29	97.99±0.26	94.32±1.57	89.97±2.08	97.98±0.34	**98.09±0.36**
Kappa(%)	86.04±1.39	96.38±0.94	95.39±0.85	96.75±0.45	96.56±0.31	92.50±1.17	89.85±1.49	96.69±0.36	**98.21±0.25**

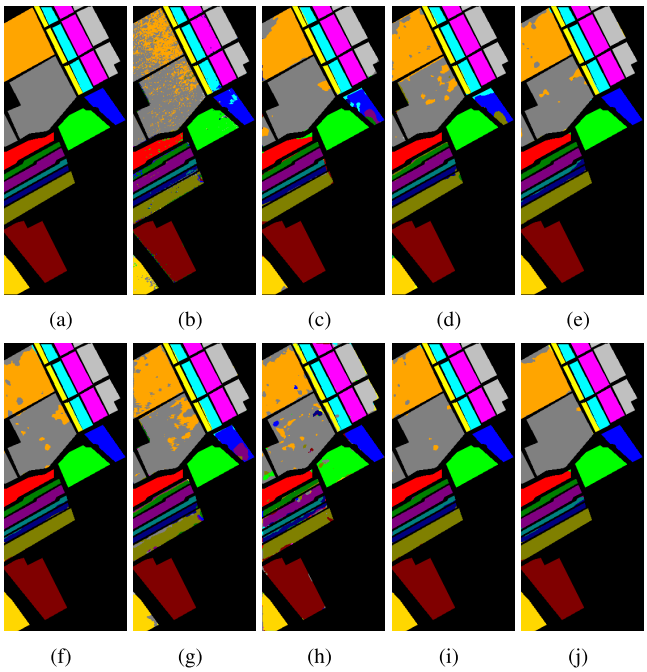


Fig. 10. Classification maps of the SA dataset obtained by different methods. (a) Ground truth. (b) SVM. (c) HybridSN. (d) SPRN. (e) HMSSF. (f) MASSFormer. (g) GSC-VIT. (h) 3DSS-Mamba. (i) IGroupSS-Mamba. (j) LKMA.

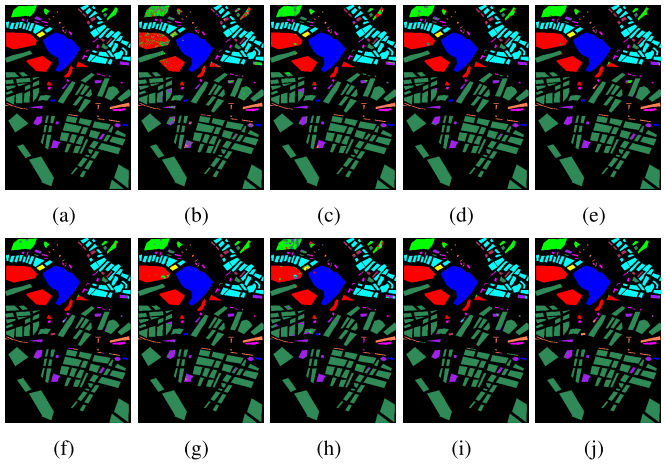


Fig. 11. Classification maps of the Tea Farm dataset obtained by different methods. (a) Ground truth. (b) SVM. (c) HybridSN. (d) SPRN. (e) HMSSF. (f) MASSFormer. (g) GSC-VIT. (h) 3DSS-Mamba. (i) IGroupSS-Mamba. (j) LKMA.

dependency modeling, and deep interaction between spatial and spectral dimensions, outperforming other compared methods. Transformer-based architectures, such as MASSFormer, are limited by secondary complexity and their lack of spatial–spectral interaction, reducing the accuracy of OA by 1.49%, AA by 0.10%, and Kappa by 1.65% compared to LKMA. In addition, the Mamba architecture relies on a single state-space model and exhibits some limitations in modeling complex scenarios. Compared with the second-best IGroupSS-Mamba, the proposed method achieves improvements of 1.11%, 0.11%, and 1.22% across OA, AA, and Kappa. Taken together, LKMA shows stronger semantic discriminative ability, which is mainly due to the comprehensive extraction and deep interaction of spatial–spectral information and further proves the superiority of the method.

3) Tea Farm Dataset: The experiments on this dataset are conducted using 1% of the available labeled samples for training. The corresponding quantitative classification results and maps are shown in Table III and Fig. 11. Due to the limited capability of local feature extraction, traditional and

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TABLE III
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE TEA FARM DATASET USING 1% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods		LKMA
	SVM	HybridSN	SPRN	HMSSF	MASSFormer	GSC-VIT	3DSS-Mamba	IGroupSS-Mamba	
C1	90.84	98.84	99.25	99.36	99.53	98.66	93.94	99.55	**99.86**
C2	31.69	88.03	92.55	97.19	96.78	89.15	53.18	98.75	**99.08**
C3	97.43	98.78	99.58	99.76	99.77	99.28	97.27	99.69	**99.85**
C4	18.07	64.10	97.90	93.30	90.71	94.10	33.11	**99.76**	90.28
C5	98.51	99.82	99.79	99.97	**100.00**	99.77	98.55	**100.00**	**100.00**
C6	87.16	81.59	92.98	91.66	89.84	**93.48**	72.89	90.63	92.05
C7	78.66	87.02	99.83	**99.93**	98.40	92.60	50.89	99.91	97.72
C8	43.53	73.39	84.16	88.75	89.51	83.04	62.54	91.31	**94.79**
C9	99.55	**99.88**	99.65	99.62	99.49	99.62	98.28	99.10	99.52
C10	90.34	95.80	97.72	88.00	96.73	98.40	80.33	92.46	**99.07**
OA(%)	91.65±0.42	97.16±0.42	98.61±0.27	98.89±0.15	99.01±0.15	98.14±0.37	92.80±4.49	99.06±0.13	**99.43±0.07**
AA(%)	73.58±2.93	88.72±3.04	96.34±0.84	95.75±0.72	96.08±0.86	94.81±1.63	74.10±14.16	97.12±0.67	**97.22±0.98**
Kappa(%)	87.47±0.65	95.78±0.62	97.93±0.40	98.35±0.23	98.53±0.22	97.24±0.55	89.23±6.69	98.61±0.20	**99.16±0.11**

TABLE IV
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE HANCHUAN DATASET USING 1% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods		LKMA
	SVM	HybridSN	SPRN	HMSSF	MASSFormer	GSC-VIT	3DSS-Mamba	IGroupSS-Mamba	
C1	94.58	98.84	99.00	99.16	98.45	97.44	96.53	99.02	**99.75**
C2	77.20	**97.54**	97.31	96.65	95.72	95.01	89.63	95.13	96.51
C3	71.47	95.67	99.18	96.94	97.04	97.03	89.83	95.19	**99.34**
C4	90.95	98.52	97.78	99.03	**99.27**	97.22	95.47	98.61	98.80
C5	17.85	80.71	94.06	90.24	92.53	89.91	76.99	87.96	**95.36**
C6	20.51	80.59	81.98	86.29	81.45	72.24	61.35	83.31	**88.83**
C7	78.77	**95.30**	93.47	93.07	87.60	88.29	88.30	92.45	93.45
C8	69.92	92.11	94.58	**96.90**	94.57	90.44	86.81	96.21	96.88
C9	60.93	93.31	95.94	**96.83**	95.73	92.87	77.82	96.76	96.14
C10	89.45	98.12	98.65	**99.20**	99.10	97.48	97.25	99.12	97.76
C11	94.66	97.05	**98.44**	98.37	96.45	95.22	95.15	98.37	98.04
C12	21.16	71.99	83.68	93.05	91.16	83.22	83.34	93.39	**94.96**
C13	47.94	85.48	85.54	85.62	83.54	82.35	71.24	85.08	**88.28**
C14	82.02	94.60	95.63	96.95	94.63	93.12	89.69	95.43	**97.63**
C15	51.88	55.02	**84.38**	78.20	71.96	82.12	70.76	77.40	82.15
C16	98.42	99.75	99.65	**99.88**	99.79	99.63	99.73	99.86	99.85
OA(%)	84.19±0.22	96.20±0.70	97.24±0.13	97.54±0.11	96.49±0.59	95.24±0.56	92.55±1.07	97.04±0.27	**97.87±0.22**
AA(%)	66.73±0.59	89.66±3.21	94.33±0.46	94.15±0.41	92.44±1.03	90.85±1.30	85.62±2.04	93.33±0.46	**95.23±0.55**
Kappa(%)	81.44±0.26	95.55±0.82	96.77±0.15	97.12±0.13	95.89±0.69	94.43±0.65	91.27±1.25	96.53±0.31	**97.50±0.26**

CNN-based methods show relatively weak results, and the proposed method LKMA improves 0.37%, 0.10%, and 0.55% for OA, AA, and Kappa compared to the next best ranked IGroupSS-Mamba. Notably, LKMA attains the highest accuracy in six categories among all compared methods. Fig. 11 illustrates the visual comparison of all approaches, where LKMA demonstrates superior smoothness. These experimental findings further confirm the effectiveness and superiority of the proposed methodology.

4) HanChuan Dataset: The experiments on this dataset are conducted using 1% of the available labeled samples for training. The corresponding quantitative classification results and maps are shown in Table IV and Fig. 12. The results show that the proposed LKMA achieves 97.87%, 95.23%, and 97.50% of OA, AA, and Kappa. Despite the high complexity of this scenario, the proposed method shows excellent advantages by virtue of the fusion of SSF and collaborative modeling with information from multiple semantics. Limited by the sensory field, the classical HybridSN method produced degraded results, compared to which OA was reduced by 1.11%, AA by 5.67%, and Kappa by 1.95%. As visualized in Fig. 12, the proposed LKMA method provided the most accurate classification results.

E. Ablation Study

1) Ablation Study on MSSFG: This section evaluates the impact of the MSSFG module on classification performance across four datasets, with OA results shown in Table V. Removing the MSSFG module leads to decreased OA on

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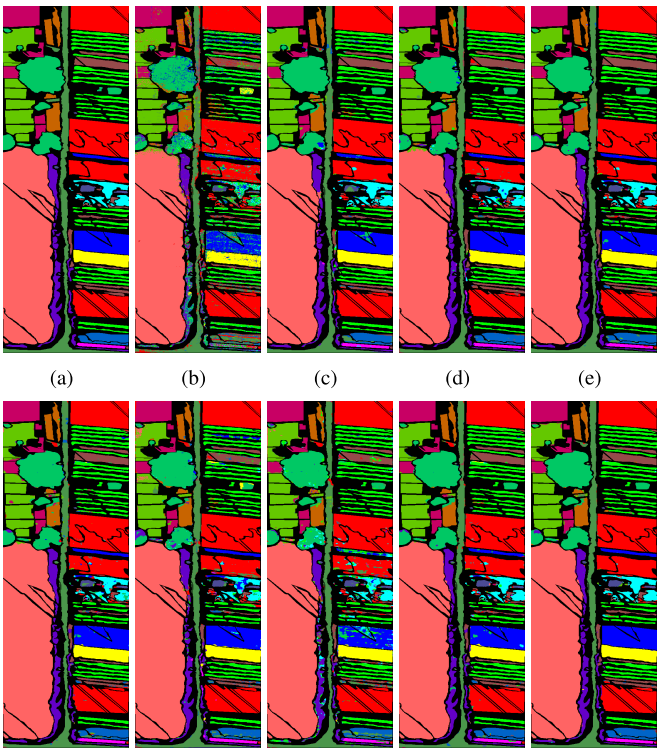


Fig. 12. Classification maps of the HanChuan dataset obtained by different methods. (a) Ground truth. (b) SVM. (c) HybridSN. (d) SPRN. (e) HMSSF. (f) MASSFormer. (g) GSC-VIT. (h) 3DSS-Mamba. (i) IGroupSS-Mamba. (j) LKMA.

TABLE V
ABLATION STUDY OF THE MSSFG MODULE
(BEST RESULTS IN BOLD)

MSSFG	The overall accuracy of different datasets			
	PU	SA	Tea Farm	HanChuan
×	97.87	98.26	99.32	97.45
√	**98.07**	**98.40**	**99.43**	**97.87**

all datasets, mainly due to the limited capability of single-scale feature extraction. By contrast, using the full MSSFG module enables multiscale feature learning, enhancing the discriminative power of the model and improving classification results. These findings highlight the crucial role of MSSFG in capturing rich spatial–spectral information for HSIC.

2) Ablation Study on LDC: The core advantage of LDC is the extraction of spatially salient features. To demonstrate its effectiveness, we compare the results of LDC with standard convolution while keeping other parameters constant. Table VI shows its OA on the four datasets. The ablation results show that the effectiveness of the module using LDC is better than the case of using standard convolution, which is mainly because of the fact that the learnable kernel mechanism in it can capture the spatial detail information more acutely and strengthen the extraction of spatial features.

3) Ablation Study on GHMM: GHMM mines global dependencies with linear complexity through a hidden state mixing mechanism. To evaluate the effectiveness of the module, we performed experiments while maintaining other parameters unchanged. Table VI shows its OA on the four datasets. Specifically, removing the GHMM model makes it difficult to capture spectral associations and spatial contextual relationships due to the lack of global semantic aggregation capability. Enabling GHMM enhances the capability to capture long-range dependencies, validating its crucial role in global semantic modeling.

TABLE VI
ABLATION STUDY OF THE KEY COMPONENTS OF OUR METHOD
(BEST RESULTS IN BOLD)

LDC	GHMM	The overall accuracy of different datasets			
		PU	SA	Tea Farm	HanChuan
×	×	94.12	98.06	99.01	96.01
√	×	96.10	98.33	99.25	97.20
×	√	97.59	98.25	99.28	97.42
√	√	**98.07**	**98.40**	**99.43**	**97.87**

TABLE VII
ABLATION STUDY OF THE KEY COMPONENTS OF OUR METHOD
(BEST RESULTS IN BOLD)

Spa	Spe	EWS	SSAF	The overall accuracy of different datasets			
				PU	SA	Tea Farm	HanChuan
√	×	×	×	91.66	96.69	97.38	96.46
×	√	×	×	96.54	97.38	98.89	96.74
√	√	√	×	97.29	97.89	99.06	97.54
√	√	×	√	**98.07**	**98.40**	**99.43**	**97.87**

4) Ablation Study on SSAF: In this part, we evaluate the influence of the spatial branch (Spa), the spectral branch (Spe), the simple elementwise summation (EWS) fusion strategy, and the SSAF module on the classification results. Table VII presents the results on the four datasets. It is clearly observed that using the simple EWS fusion can moderately improve the classification performance compared to using Spa or Spe individually. However, when the features are fused through the SSAF module, the classification performance is further significantly enhanced, surpassing both EWS fusion and individual branches. This advancement is achieved through the SSAF module's capability to integrate semantic information across feature groups via SGSA and PSSA, thereby enhancing spatial–spectral interactions and representation learning. These results also highlight the effectiveness and advantages of the SSAF module.

F. Robustness Assessment

To evaluate the robustness of the proposed LKMA method, extensive experiments are conducted on the PU, SA, Tea Farm, and HanChuan datasets by varying the proportions of training samples. Fig. 13 presents the results for all comparison methods, with the performance of LKMA prominently highlighted by red curves. The training sample ratios are set between 0.1% and 1% for the PU and SA datasets, and between 0.2% and 2% for the Tea Farm and HanChuan datasets. The results indicate that the classification accuracy of LKMA consistently

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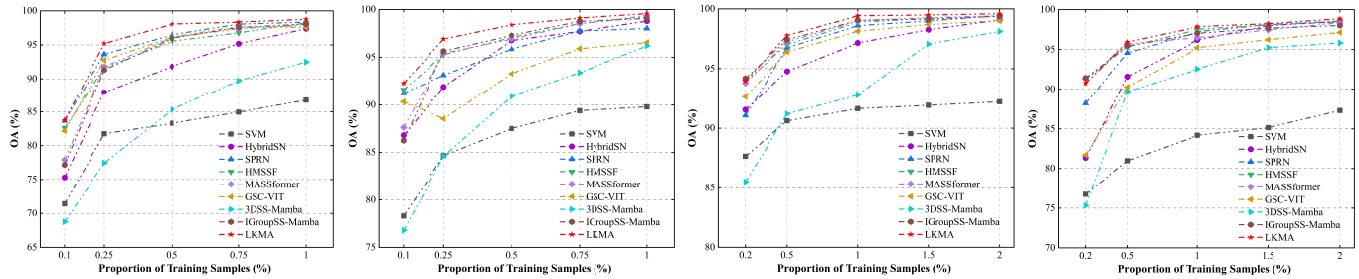


Fig. 13. Classification performance under varying training sample proportions on four datasets. (a) PU. (b) SA. (c) Tea Farm. (d) HanChuan.

improves as the training sample ratio increases. Moreover, LKMA outperforms other comparison methods across most sampling ratios, further demonstrating the effectiveness and robustness of the proposed approach.

V. Conclusion

This article introduces a novel LKMA framework for HSIC, which progressively extracts and integrates local and global SSF while enhancing edge information representation. The MSSFG module employs multiscale LDCs and multiscale dilation convolutions to capture spatial semantic information and spectral correlations, respectively. The GHMM module utilizes a hidden state mixing mechanism for modeling long-range dependencies across spatial and spectral dimensions. The SSAF module combines SGSA and PSSA mechanisms to explore the synergistic effects between spatial and spectral attention, thereby enhancing deep interactions of SSF. Our experiments demonstrate that the proposed method, LKMA, provides an effective solution for fine feature recognition in complex scenes relative to state-of-the-art methods.

The LKMA presents notable benefits in HSIC, but its parameter tuning is empirically dependent and affects the generalization efficiency. Future research will focus on automated parameter optimization and lightweight design to improve the applicability of the model for real-time processing and resource-constrained scenarios.

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